

Chapter 1

Introduction

At least since the risk management efforts to assess credit default swaps by banks leading up to the 2008 financial crisis the interest of hedging derivatives has arised. Especially, tail risks and rare events, not following a normal distribution, have become a topic in finance. The aim of this thesis is to provide a framework for hedging these risks in a model-free way.

In a frictionless complete market every claim can be replicated perfectly and the replication cost is its unique arbitrage-free price; see Föllmer and Schied [4, Chapter 5]. Real markets have transaction costs and trading constraints making perfect replication impossible. The hedger thus retains residual risk. This thesis ranks profit-and-loss outcomes by convex risk measures [4, Chapter 4] and views hedging as a trading strategy that minimizes the risk measure.

For this, Bühler, Gonon, Teichmann and Wood [2] proposed *deep hedging*. On a *finite* probability space their Proposition 4.3 shows that the minimal risk π_M of deep neural network strategies of size M decreases to the true minimal risk π .

This thesis develops the framework rigorously on a general probability space, which turns two questions that are trivial on finite Ω into genuine analytical problems. The first is attainment: is the infimum defining the hedging functional attained by an optimal strategy? The second is approximation: do Neural Networks approximate the optimal hedge? We show that the two questions fail on general Ω , and then prove that for the optimized certainty equivalents of Ben-Tal and Teboulle [1] the results are restored under further imposed economically sensible assumptions.

Chapter 2 constructs the market and the risk measures, Chapter 3 proves the attainment theorem, Chapter 4 the approximation results and Chapter 5 concludes; the appendix proves the Komlós lemma; for backtests see [2].

Bibliography

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- [3] F. Delbaen and W. Schachermayer. A general version of the fundamental theorem of asset pricing. *Mathematische Annalen* **300**(1), 463–520, 1994.
- [4] H. Föllmer and A. Schied. *Stochastic Finance: An Introduction in Discrete Time*. 4th edition, De Gruyter, Berlin, 2016.